Computer Vision Fundamentals

What Is Computer Vision?

Computer vision is the subfield of artificial intelligence concerned with enabling computers to extract meaningful information from digital images and video, and to act on or reason about that information. Just as natural language processing deals with the challenge of understanding text, computer vision deals with the challenge of understanding pixels: raw grids of numbers representing color intensity that, on their own, carry no explicit meaning to a computer. The goal of computer vision is to bridge this gap, allowing a machine to answer questions such as what objects are present in an image, where they are located, what is happening in a video, or whether an image matches a particular pattern of interest, such as a defect on a manufacturing line or a tumor in a medical scan.

Core Computer Vision Tasks

Several standard tasks define the field. Image classification assigns a single label to an entire image, such as determining whether a photo contains a cat or a dog. Object detection goes a step further by identifying multiple objects within an image and drawing a bounding box around each one, along with a label, which is essential for applications such as autonomous driving, where a system must locate pedestrians, vehicles, and traffic signs simultaneously. Semantic segmentation assigns a label to every individual pixel in an image, grouping pixels into regions such as "road," "sky," or "building," which provides a much finer-grained understanding of a scene than bounding boxes alone. Instance segmentation combines object detection and segmentation by both identifying separate object instances and delineating their exact pixel boundaries. Facial recognition, a widely deployed but also controversial application, identifies or verifies a person's identity based on facial features. Optical character recognition, commonly abbreviated OCR, converts images of printed or handwritten text into machine-readable text, and is used in applications ranging from digitizing scanned documents to reading license plates.

From Hand-Crafted Features to Deep Learning

Before the deep learning era, computer vision systems relied heavily on hand-crafted feature extraction algorithms designed by researchers, such as edge detectors, corner detectors, and descriptors like SIFT and HOG, which converted raw pixels into a smaller set of numerical features believed to be relevant for a given task. These hand-crafted features were then fed into traditional machine learning classifiers, such as support vector machines, to make a final prediction. While clever, these hand-crafted approaches struggled to generalize across the enormous variability of real-world images, including changes in lighting, viewpoint, and occlusion.

The shift to deep learning, and specifically to convolutional neural networks as described in the deep learning material in this document collection, transformed the field starting around 2012, when a deep convolutional network dramatically outperformed all competing approaches on the large-scale ImageNet image classification benchmark. Since that breakthrough, deep learning has become the standard approach for nearly all computer vision tasks, because convolutional networks can learn the relevant visual features directly from data rather than requiring them to be manually designed.

Convolutional Neural Networks for Vision

As introduced in the deep learning overview, convolutional neural networks apply small, learnable filters across an image to detect local visual patterns such as edges, textures, and shapes, and then combine these patterns hierarchically across many layers to recognize increasingly complex structures, from simple edges to entire objects. Architectures built specifically for vision, such as ResNet, which introduced residual connections to allow training of very deep networks, and later vision transformers, which adapt the self-attention mechanism originally developed for natural language processing to work on patches of an image, have driven continuous improvements in accuracy on standard vision benchmarks. Modern object detection systems, such as the YOLO family of models, are able to process a full image in a single pass through a neural network and are fast enough to run in real time, which is essential for applications such as video surveillance and autonomous vehicles.

Training Data and Data Augmentation in Computer Vision

Computer vision models are especially data-hungry, since visual concepts can appear under an enormous range of variation in lighting, angle, scale, occlusion, and background. Large labeled datasets, such as ImageNet, COCO, and many domain-specific medical imaging datasets, have been essential for training accurate models. Because collecting and labeling images is expensive and time-consuming, practitioners rely heavily on data augmentation, which artificially expands a training set by applying transformations such as rotation, flipping, cropping, and color adjustment to existing images, teaching the model to be robust to these variations without requiring additional labeled examples. Transfer learning, in which a model is first pretrained on a very large general-purpose dataset like ImageNet and then fine-tuned on a smaller, task-specific dataset, is also widely used, allowing practitioners to achieve strong performance even when they only have a limited number of labeled examples for their specific application.

Real-World Applications of Computer Vision

Computer vision technology is deployed across many industries. In healthcare, computer vision systems assist radiologists by flagging suspicious regions in X-rays, CT scans, and MRI images. In manufacturing, vision systems perform automated quality inspection, detecting defects on production lines faster and more consistently than human inspectors. In retail, vision systems

power cashier-less checkout systems and inventory tracking. In agriculture, vision systems mounted on drones or tractors monitor crop health and identify pests. In transportation, computer vision is a core component of advanced driver-assistance systems and autonomous vehicles, where it must fuse information about lanes, other vehicles, pedestrians, and traffic signs, often in combination with other sensors such as radar and lidar.

Limitations and Challenges

Despite significant progress, computer vision systems still face important challenges. Models can be fooled by adversarial examples, which are images subtly modified in ways imperceptible to humans but that cause a model to misclassify them entirely. Models trained on data that does not represent the full diversity of real-world conditions, for example a facial recognition system trained mostly on one demographic group, can exhibit significant accuracy disparities across different groups, raising serious fairness and safety concerns, particularly for applications like facial recognition. Vision systems can also struggle with scenes very different from their training data, such as unusual weather conditions for an autonomous vehicle, and typically cannot explain their reasoning in a way that is easy for a human to audit, similar to the interpretability challenges described for deep learning models more broadly.